

```
[root@vbg ~]# head -n 10 /etc/openldap/slapd.conf
#
# See slapd.conf(5) for details on configuration options.
# This file should NOT be world readable.
#
include      /etc/openldap/schema/core.schema
include      /etc/openldap/schema/cosine.schema
include      /etc/openldap/schema/inetorgperson.schema
include      /etc/openldap/schema/nis.schema
```

The 'include' directive in *slapd.conf* reads information from other files. The schema files are generally included using this directive. Each schema file can be understood to be analogous to a spreadsheet file. Each schema file defines certain ObjectClasses and attributes. Continuing with our analogy, ObjectClasses can be considered to be similar to each sheet within a spreadsheet file. Attributes are then analogous to each column within a sheet.

In other words, multiple attributes are grouped together in an ObjectClass. Multiple ObjectClasses can be grouped to form a record.

The first schema file included in the *slapd.conf* configuration file, by default, is *core.schema*. This file defines the 'core' schema of OpenLDAP. Going by the above analogy, the ObjectClasses defined in this file are:

```
[root@vbg ~]# cat /etc/openldap/schema/core.schema | egrep ^objectclass
objectclass ( 2.5.6.2 NAME 'country'
objectclass ( 2.5.6.3 NAME 'locality'
objectclass ( 2.5.6.4 NAME 'organization'
objectclass ( 2.5.6.5 NAME 'organizationalUnit'
objectclass ( 2.5.6.6 NAME 'person'
objectclass ( 2.5.6.7 NAME 'organizationalPerson'
objectclass ( 2.5.6.8 NAME 'organizationalRole'
objectclass ( 2.5.6.9 NAME 'groupOfNames'
objectclass ( 2.5.6.10 NAME 'residentialPerson'
objectclass ( 2.5.6.11 NAME 'applicationProcess'
objectclass ( 2.5.6.12 NAME 'applicationEntity'
objectclass ( 2.5.6.13 NAME 'dsa'
objectclass ( 2.5.6.14 NAME 'device'
objectclass ( 2.5.6.15 NAME 'strongAuthenticationUser'
objectclass ( 2.5.6.16 NAME 'certificationAuthority'
objectclass ( 2.5.6.17 NAME 'groupOfUniqueNames'
objectclass ( 2.5.6.18 NAME 'userSecurityInformation'
objectclass ( 2.5.6.16.2 NAME 'certificationAuthority-V2'
objectclass ( 2.5.6.19 NAME 'cRLDistributionPoint'
objectclass ( 2.5.6.20 NAME 'dmd'
objectclass ( 2.5.6.21 NAME 'pkouser'
objectclass ( 2.5.6.22 NAME 'pkica'
objectclass ( 2.5.6.23 NAME 'deltaCRL'
objectclass ( 1.3.6.1.4.1.250.3.15 NAME 'labeledURIObject'
objectclass ( 0.9.2342.19200300.100.4.19 NAME 'simpleSecurityObject'
objectclass ( 1.3.6.1.4.1.1466.344 NAME 'dcObject'
objectclass ( 1.3.6.1.1.3.1 NAME 'uidObject'
```

In other words, our spreadsheet file, called *core.schema*, has 27 sheets called 'country', 'locality', and so on.

Let us now look closely at the definition of our sheet called 'country':

```
[root@vbg ~]# cat /etc/openldap/schema/core.schema | \
grep -A 5 "objectclass ( 2.5.6.2 NAME 'country'"
objectclass ( 2.5.6.2 NAME 'country'
DESC 'RFC2256: a country'
SUP top STRUCTURAL
MUST c
MAY ( searchGuide $ description ) }
```

Let us now look for the columns in this sheet called 'country'. These columns can be found in the 'MUST' and the 'MAY' directives within the ObjectClass definition. Going by this, the 'columns' in the 'country' sheet are:

c, searchGuide and description

The next question that naturally arises is: what are 'c', 'searchGuide' and 'description'? Going by the spreadsheet analogy, can we fill in numbers, date, characters as data into these columns? What kind of data can we put into these columns? This is also defined in the schema file:

```
[root@vbg ~]# cat /etc/openldap/schema/core.schema | \
grep ^attribute | grep searchGuide
attributetype ( 2.5.4.14 NAME 'searchGuide'

[root@vbg ~]# cat /etc/openldap/schema/core.schema | \
grep -A 3 "attributetype ( 2.5.4.14 NAME 'searchGuide'"
attributetype ( 2.5.4.14 NAME 'searchGuide'
DESC 'RFC2256: search guide, deprecated by enhancedSearchGuide'
SYNTAX 1.3.6.1.4.1.1466.115.121.1.25 }
```

The above lines contain the definition for the attribute 'searchGuide'. The important things to note are:

- **attributetype (2.5.4.14 NAME 'searchGuide'** – This specifies that a new attribute definition follows. The name (NAME) of the attribute is searchGuide and its OID number is 2.5.4.14. OID are Object Identifiers assigned by IANA. We will discuss these in more detail later. At this point, it is sufficient to understand the following:
 - **OID 2 =>** This refers to ISO/ITU-T jointly assigned OIDs
 - **OID 2.5 =>** This refers to X.500 Directory Services
 - **OID 2.5.4 =>** This refers to X.500 attribute types, and
 - **OID 2.5.4.14 =>** This refers to the attribute 'searchGuide'. The searchGuide attribute type specifies information of suggested search criteria, which may be included in some entries expected to be a convenient base-object for the search operation—for example, country or organisation. You can find more details at www.alvestrand.no/objectid/top.html. Do visit this URL to understand in greater detail how OIDs are assigned and what OIDs have been assigned to whom.
- **DESC 'RFC2256: search guide, deprecated by enhancedSearchGuide'** – This gives a description for the